CSE-476: Introduction to Natural Language Processing
Quiz 1

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

Q1. In the BIO scheme for NER, what does the "I-" tag represent?

A. Beginning of an entity

B. Inside an entity

C. Outside any entity

D. Between two tokens

Q2. In sequence tagging, which model type uses only words and ignores labels entirely?

A. It uses only words (observations) and ignores labels

B. It is identical to an HMM

C. It models transitions between labels but does not consider the words

D. It models emissions P(word | label)

Q3. In an HMM for NER, which probability represents transitions between labels?

A. P(label | previous label)

B. P(label | word)

C. P(word | previous word)

D. P(word | label)

Q4. Which of the following assigns grammatical roles like subject and object from syntax?

A. A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location

B. A process that assigns grammatical roles like subject and object based on syntactic structure

C. A method that classifies named entities and links them to a knowledge base

D. A technique that labels words with their parts of speech and morphological features

Q5. Which technique compresses text by replacing repeated characters with a count and symbol?

A. A technique that compresses text by replacing consecutive repeated characters with a count and a symbol

B. A tokenization method that iteratively merges the most frequent pair of symbols in a corpus

C. A tokenization method that splits text based on whitespace and punctuation

D. A character-level encoding scheme that assigns fixed-length binary codes to characters

Q6. Which task labels each token in a sentence with a grammatical category?

A. To label each token with a grammatical category

B. To convert text into a sequence of symbols suitable for modeling

C. To reduce vocabulary size to zero

D. To assign probabilities to sentences

Q7. What is a drawback of a tokenizer that ignores punctuation characters entirely?

A. It produces too many tokens

B. It does not work well for languages without explicit word boundaries

C. It cannot tokenize English at all

D. It ignores punctuation entirely

Q8. Which of the following is commonly treated as a sequence tagging problem?

A. Part-of-speech tagging

B. Named Entity Recognition

C. Dependency parsing

D. Sentiment analysis

Q9. Which issue involves a tagging scheme that requires an excessively large number of labels?

A. It cannot represent overlapping or nested entities

B. It requires too many labels

C. It does not support multiple entity types

D. It ignores entity boundaries

Q10. In model selection, which trade-off refers to balancing system speed a

A. Speed vs accuracy

B. Tokenized sequence length vs vocabulary size

C. Precision vs recall

D. Memory vs computation